

Prop transformada fourier traslacion modulacion dilatacion

Proposición 1. Sea $f \in \mathcal{L}^1(\mathbb{R})$, $y, z \in \mathbb{R}, \delta > 0$, entonces

$$(1) (T_y f)^\wedge(\xi) = e^{-2\pi i y \xi} f^\wedge(\xi) = M_{-y} f^\wedge(\xi);$$

$$(2) (M_z f)^\wedge(\xi) = f^\wedge(\xi - z) = T_z(f^\wedge)(\xi);$$

$$(3) (D_\delta f)^\wedge(\xi) = \frac{1}{\delta} f^\wedge\left(\frac{\xi}{\delta}\right) = \frac{1}{\delta} D_{\frac{1}{\delta}}(f^\wedge)(\xi);$$

donde $f^\wedge$ denota la transformada de Fourier de f .

Demostración:

(1) Calculamos desde las definiciones y hacemos el cambio de variable $u = x - y$:

$$\begin{aligned} (T_y f)^\wedge(\xi) &= \int_{\mathbb{R}} f(x - y) e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} f(u) e^{-2\pi i(u+y)\xi} du \\ &= e^{-2\pi i y \xi} \int_{\mathbb{R}} f(u) e^{-2\pi i u \xi} du = e^{-2\pi i y \xi} f^\wedge(\xi) = M_{-y} f^\wedge(\xi). \end{aligned}$$

(2) En este caso, no es necesario un cambio de variable:

$$(M_z f)^\wedge(\xi) = \int_{\mathbb{R}} e^{2\pi i z x} f(x) e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} f(x) e^{-2\pi i x(\xi - z)} dx = f^\wedge(\xi - z) = T_z(f^\wedge)(\xi).$$

(3) Hacemos el cambio de variable $u = \delta x$:

$$\begin{aligned} (D_\delta f)^\wedge(\xi) &= \int_{\mathbb{R}} f(\delta x) e^{-2\pi i x \xi} dx = \int_{\mathbb{R}} f(u) e^{-2\pi i \frac{u}{\delta} \xi} \frac{du}{\delta} \\ &= \frac{1}{\delta} \int_{\mathbb{R}} f(u) e^{-2\pi i u \frac{\xi}{\delta}} du = \frac{1}{\delta} f^\wedge\left(\frac{\xi}{\delta}\right) = \frac{1}{\delta} D_{\frac{1}{\delta}}(f^\wedge)(\xi). \end{aligned}$$

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